

Case Study

Assessment week 2 | Case Study 1 Employee Performance and Attrition

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(Batch 3)



https://github.com/amankumar1ext-web/Case_Study/tree/main/W2_Case_Study_1

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Employee Performance and Attrition Analysis project focuses on analyzing employee-related data to understand workforce trends, employee performance, job satisfaction and attrition. The project combines Python, MySQL/SQL and Power BI to create a structured analytics process starting from raw employee datasets and ending with an interactive business dashboard.

The project uses three datasets: Employee Data, Attrition Data and Employee Performance Data. Employee Data contains demographic and job-related information, Attrition Data contains employee attrition and exit-interview information, and Employee Performance Data contains performance ratings, promotion information, training hours, work-life balance and job satisfaction. These datasets are analyzed together using the common employee identifier.

The implementation begins with Python-based data loading, inspection and preprocessing. The cleaned data is then organized and stored in MySQL, where SQL is used for data management, transformation, querying and analytical processing. The processed data is finally connected to Power BI, where KPIs, charts, filters and interactive reports are created.

The overall purpose is to transform raw employee information into meaningful analytical information that can help understand employee performance and attrition patterns. The case study specifically focuses on factors such as salary, work-life balance and tenure, while also considering performance, satisfaction and department-level trends.

1.2 Problem Statement

Organizations face challenges in managing employee performance and retaining employees. High employee attrition can increase hiring costs, result in the loss of organizational knowledge and affect productivity and business operations.

The case study identifies difficulties in accurately assessing employee performance trends, identifying factors associated with voluntary and involuntary

attrition, and understanding relationships between compensation, job satisfaction, work-life balance and employee retention.

Therefore, this project implements a data analytics pipeline to organize employee data, perform analysis and present the resulting information through an interactive Power BI dashboard. The analysis focuses on measurable employee and organizational attributes rather than relying only on manual HR reporting.

1.3 Project Objectives

The main objectives of the project are:

- To identify factors associated with employee attrition.
- To analyze employee performance and satisfaction.
- To examine employee tenure and department-level trends.
- To prepare and transform employee datasets for analysis.
- To create a structured SQL data model.
- To perform analytical queries using SQL.
- To create interactive Power BI reports.
- To calculate and display important HR KPIs.
- To support data-driven HR analysis and decision-making.

1.4 Scope of the Project

The scope of this project covers the complete analytical workflow from raw datasets to dashboard reporting.

The implementation includes:

- Loading the three provided datasets.
- Understanding dataset structure and relationships.
- Cleaning and preprocessing data using Python.
- Performing exploratory data analysis using Python.

- Creating and managing tables in MySQL.
- Establishing relationships between relevant datasets.
- Performing data transformation and analytical SQL queries.
- Connecting the processed data with Power BI.
- Creating DAX-based measures and KPIs.
- Developing an interactive employee performance and attrition dashboard.
- Presenting analytical findings through visualizations.

The project focuses on the data and attributes provided in the case study rather than implementing a separate machine-learning prediction model.

1.5 Technology Stack

Technology	Purpose
Python	Data loading, preprocessing and exploratory analysis
Jupyter Notebook	Python development and documentation
Pandas	Data manipulation and analysis
MySQL	Data storage, modeling and SQL analysis
SQL	Data transformation, querying and analytical processing
Power BI Desktop	Data modeling, KPI creation and visualization
Microsoft Word	Final project documentation

The case study identifies Data Warehouse, SQL, Python and Power BI as the main technologies for the project. Python is intended for preprocessing and statistical analysis, SQL for extraction, transformation, loading and analysis, and Power BI for visualization and reporting.

1.6 Project Implementation Methodology

The project follows a sequential data analytics pipeline. The raw CSV datasets are first loaded into Python for inspection and preprocessing. After cleaning and transformation, the data is organized in MySQL for structured storage and SQL-

based analysis. The processed data is then connected to Power BI, where the data model, DAX measures, KPIs and interactive visualizations are developed.

This approach provides a continuous flow from raw data to analytical reporting.

Project Pipeline

Raw Data → Python → MySQL → SQL Analysis → Power BI → KPIs & Dashboard → Insights

The implementation pipeline is divided into the following stages:

- Data Collection – Using the three provided CSV datasets as the source data.
- Data Preparation – inspect and clean the datasets.
- Data Modeling – Using MySQL to organize the processed data into related tables.
- SQL Analysis – Using SQL queries to transform and analyze the stored data.
- BI Modeling – Connecting the processed data to Power BI and establishing required relationships.
- KPI Development – Creating measures for employee and attrition analysis.
- Dashboard Development – Creating interactive visualizations, slicers and KPI cards.
- Insight Generation – Using the dashboard to identify relevant employee performance and attrition patterns.

1.7 Key Performance Indicators

The project dashboard will focus on the KPIs specified in the case study:

- Attrition Rate
- Retention Rate
- Average Tenure
- Department-wise Employee Score

- Average Performance Rating
- Average Exit Interview Satisfaction Score
- Department-wise Attrition Rate

These KPIs will be calculated and presented in Power BI using the prepared datasets and appropriate measures.

1.8 Expected Outcomes

The expected outcome of the project is a structured employee analytics solution that provides an understanding of employee performance and attrition patterns.

The completed solution will contain cleaned datasets, a structured MySQL data model, SQL analysis and an interactive Power BI dashboard. The dashboard will provide HR-related KPIs and visual analysis that can be used to examine employee trends and factors related to attrition.

The case study expects the solution to provide a clearer understanding of factors affecting employee performance and attrition, actionable information for HR teams and a user-friendly dashboard for continuous monitoring and decision-making.

CHAPTER 2: DATASET UNDERSTANDING & INITIAL DATA INSPECTION

2.1 Load Dataset

Loading Source Data: Using Python Pandas to load the three source datasets for further inspection and preprocessing.

```
print("Employee Data Shape:", employee_df.shape)
print("Attrition Data Shape:", attrition_df.shape)
print("Employee Performance Data Shape:", performance_df.shape)

Employee Data Shape: (1000, 12)
Attrition Data Shape: (144, 3)
Employee Performance Data Shape: (1000, 6)
```

2.2 Display Column Names

Column Identification: The column names of each dataset were inspected to understand the available employee, attrition and performance attributes.

2.3 Preview the Data

	Employee_ID	Age	first_name	last_name	gender	Gender	Department	Job_Role	Education_Level	Marital_Status	Job_Tenure	Distance_From_Home
0	1	22	Konstantin	Shutler	Male	Other	HR	Developer	PhD	Married	1	8
1	2	38	Thorvald	Skoggins	Male	Other	IT	Marketing Lead	Master's	Married	18	32
2	3	35	Gracia	Teggin	Genderqueer	Male	Sales	Senior Developer	Bachelor's	Divorced	4	42
3	4	37	Kerwinn	Romme	Male	Male	Marketing	Senior Developer	Bachelor's	Divorced	5	43
4	5	26	Randene	Millier	Female	Female	Operations	Manager	Master's	Single	10	33

Attrition Data

	employee_ID	attrition	Exit_Interview_Score
0	474	True	2
1	97	True	4
2	729	False	1
3	278	False	2
4	470	False	4

ETC.

Initial Data Preview: The first few records of each dataset were reviewed to understand the structure and values of the source data.

2.4 Check Data Types

```
Employee Performance Data Information
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 6 columns):
#   Column                      Non-Null Count  Dtype
---  -
0   Employee_ID                 1000 non-null  int64
1   Performance_Rating          1000 non-null  int64
2   Last_Promotion_Year         1000 non-null  int64
3   Training_Hours              1000 non-null  int64
4   Work_Life_Balance           1000 non-null  int64
5   Job_Satisfaction            1000 non-null  int64
dtypes: int64(6)
memory usage: 47.0 KB
```

Data Type Inspection: inspect column data types and non-null record counts before data cleaning.

2.5 Check Missing Values

```
print("\nMissing Values in Attrition Data")
print(attrition_df.isnull().sum())

print("\nMissing Values in Employee Performance Data")
print(performance_df.isnull().sum())
```

Missing Values in Employee Data

Employee_ID	0
Age	0
first_name	0
last_name	0
gender	0
Gender	0
Department	0
Job_Role	0
Education_Level	0
Marital_Status	0
Job_Tenure	0
Distance_From_Home	0

dtype: int64

Missing Values in Attrition Data

employee_ID	0
attrition	0
Exit_Interview_Score	0

dtype: int64

Missing Value Check: identify missing values in each dataset before applying data-cleaning rules.

2.6 Key takeaway

Employee Data has 1,000 records and 12 columns, with no missing values and no duplicate rows.

Attrition Data has 144 records and contains 2 duplicate rows.

Employee Performance Data has 1,000 records and 6 columns, with no missing values and no duplicate rows.

Employee Data contains both gender and Gender.

ETC

Employee ID Validation: validate the uniqueness of employee identifiers before establishing dataset relationships.

```
Unique Employee IDs: 1000
Total Employee Records: 1000

Unique Performance Employee IDs: 1000
Total Performance Records: 1000

Unique Attrition Employee IDs: 127
Total Attrition Records: 144
```

Identify Duplicate Attrition Records

2.7 Data Analysis

Check	Result	Interpretation
Employee records	1,000	Complete employee master dataset
Unique Employee IDs	1,000	Employee ID is unique
Performance records	1,000	One record per employee
Unique Performance IDs	1,000	Performance can relate 1:1 to Employee
Attrition records	144	Smaller than employee master
Unique Attrition IDs	127	Some employees have multiple attrition records
Attrition IDs missing from Employee Data	0	All attrition IDs have a matching employee
Performance IDs missing from Employee Data	0	All performance IDs have a matching employee

Employee values missing	0	No missing values detected
Performance values missing	0	No missing values detected
Attrition missing values	0	No missing values detected
Employee rows duplicate	0	No exact duplicate rows
Performance rows duplicate	0	No exact duplicate rows
Attrition duplicate rows	2	Exact duplicates exist

```
display(gender_comparison)
```

Gender	Female	Male	Other
gender			
Agender	6	9	6
Bigender	9	8	4
Female	155	145	153
Genderfluid	9	3	4
Genderqueer	5	6	5
Male	141	139	162
Non-binary	6	7	6
Polygender	5	4	3

Categorical Field Validation: inspect the consistency of gender-related categorical fields before standardization.

Data Quality Summary: Python was used to consolidate the initial record, column, missing-value, duplicate and employee-ID checks.

```
display(data_quality_summary)
```

	Dataset	Rows	Columns	Missing Values	Duplicate Rows	Unique Employee IDs
0	Employee Data	1000	12	0	0	1000
1	Attrition Data	144	3	0	2	127
2	Employee Performance Data	1000	6	0	0	1000

2.8 Perform Data Cleaning & Standardization

What I do here:

I will remove only exact duplicate records. We will not remove records merely because Employee_ID is repeated.

I will retain both source attributes but give them clearer standardized names.

```
print("Employee Data")
print("Rows:", len(employee_clean))
print("Duplicate rows:", employee_clean.duplicated().sum())

print("\nAttrition Data")
print("Rows:", len(attrition_clean))
print("Duplicate rows:", attrition_clean.duplicated().sum())

print("\nPerformance Data")
print("Rows:", len(performance_clean))
print("Duplicate rows:", performance_clean.duplicated().sum())
```

```
Employee Data
Rows: 1000
Duplicate rows: 0
```

```
Attrition Data
Rows: 142
Duplicate rows: 0
```

```
Performance Data
Rows: 1000
Duplicate rows: 0
```

Post-Cleaning Validation: Dataset dimensions and duplicate counts were rechecked after preprocessing.

```
display(final_quality_summary)
```

	Dataset	Original Rows	Final Rows	Columns	Missing Values	Duplicate Rows
0	Employee Data	1000	1000	12	0	0
1	Attrition Data	144	142	3	0	0
2	Employee Performance Data	1000	1000	6	0	0

Final Data Quality Summary after Python Preprocessing

CHAPTER 3: EXPLORATORY DATA ANALYSIS (EDA)

```
plt.show()
```



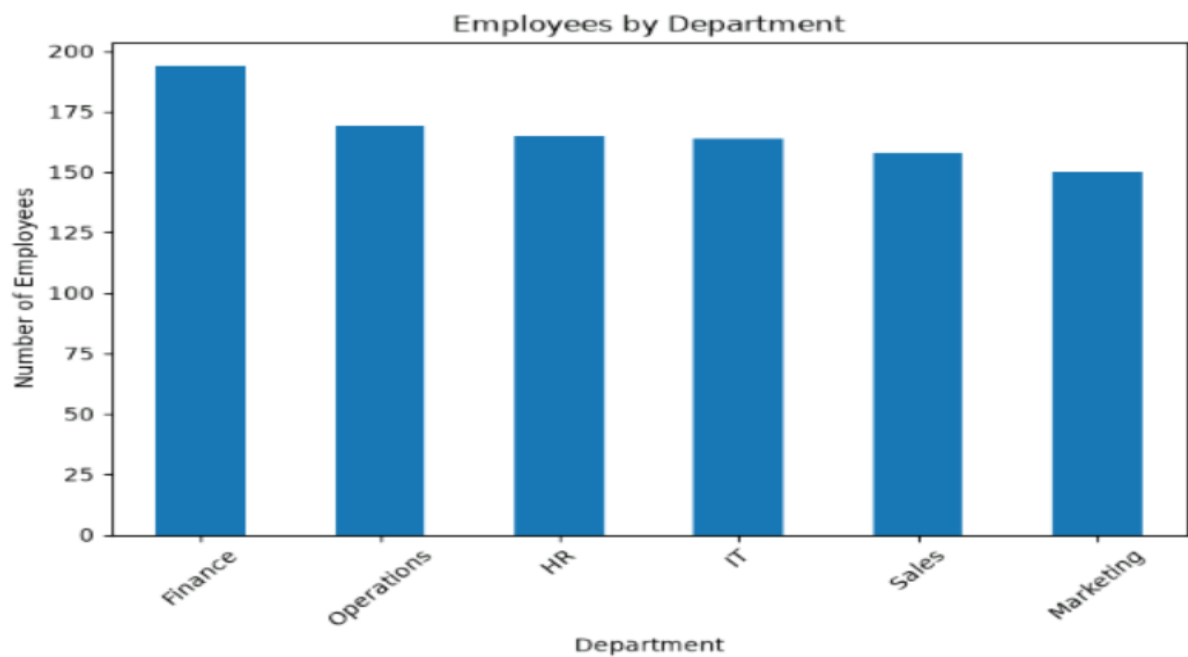
Age Distribution: visualize the distribution of employee ages.

```
plt.show()
```



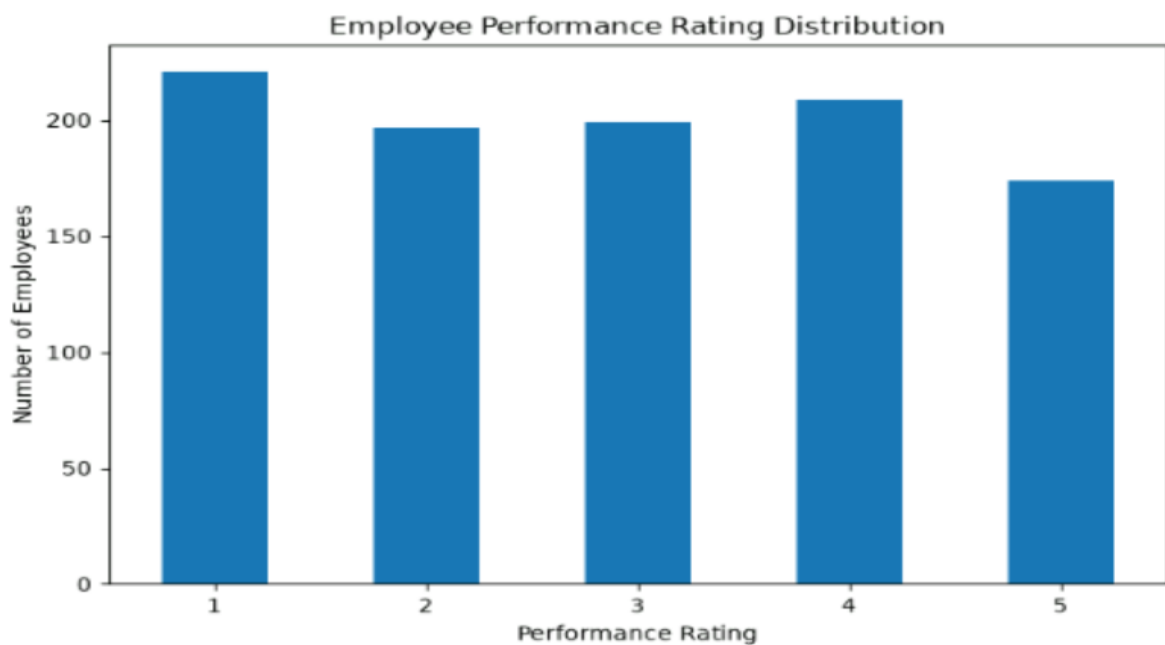
Tenure Distribution: visualize the distribution of employee tenure in the organization.

```
plt.show()
```



Department Distribution: examine employee distribution across organizational departments.

```
plt.show()
```



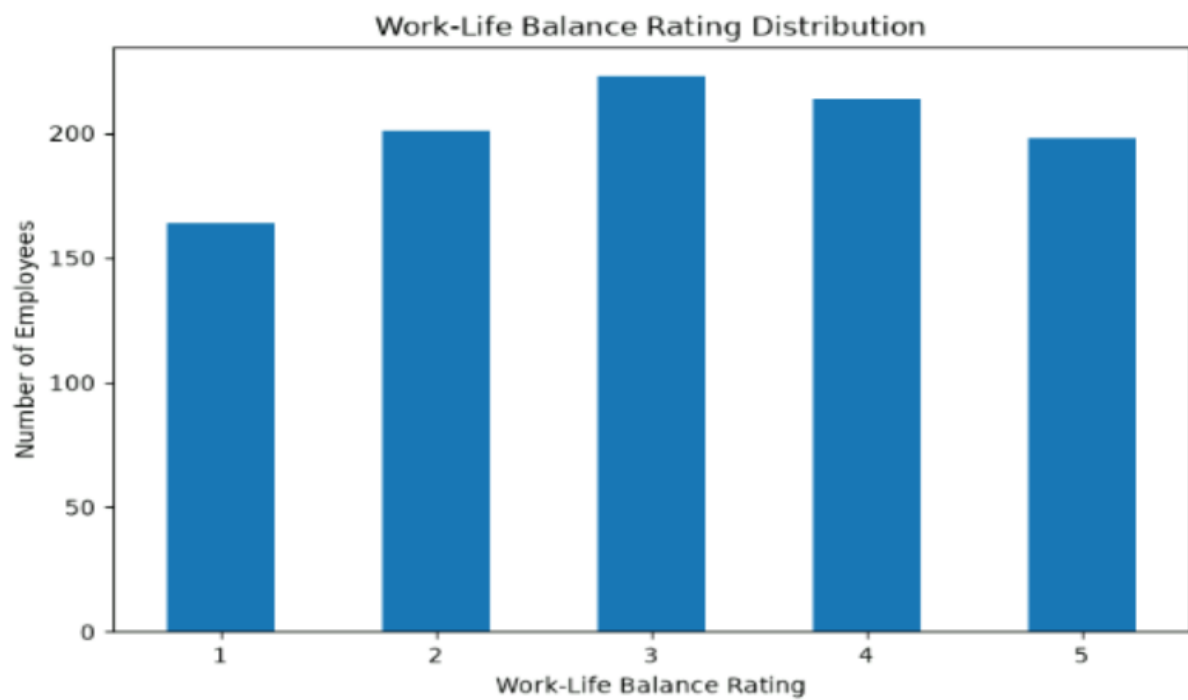
Performance Rating Distribution: examine the distribution of employee performance ratings.

```
plt.show()
```



Training Hours Distribution: examine the distribution of employee training hours.

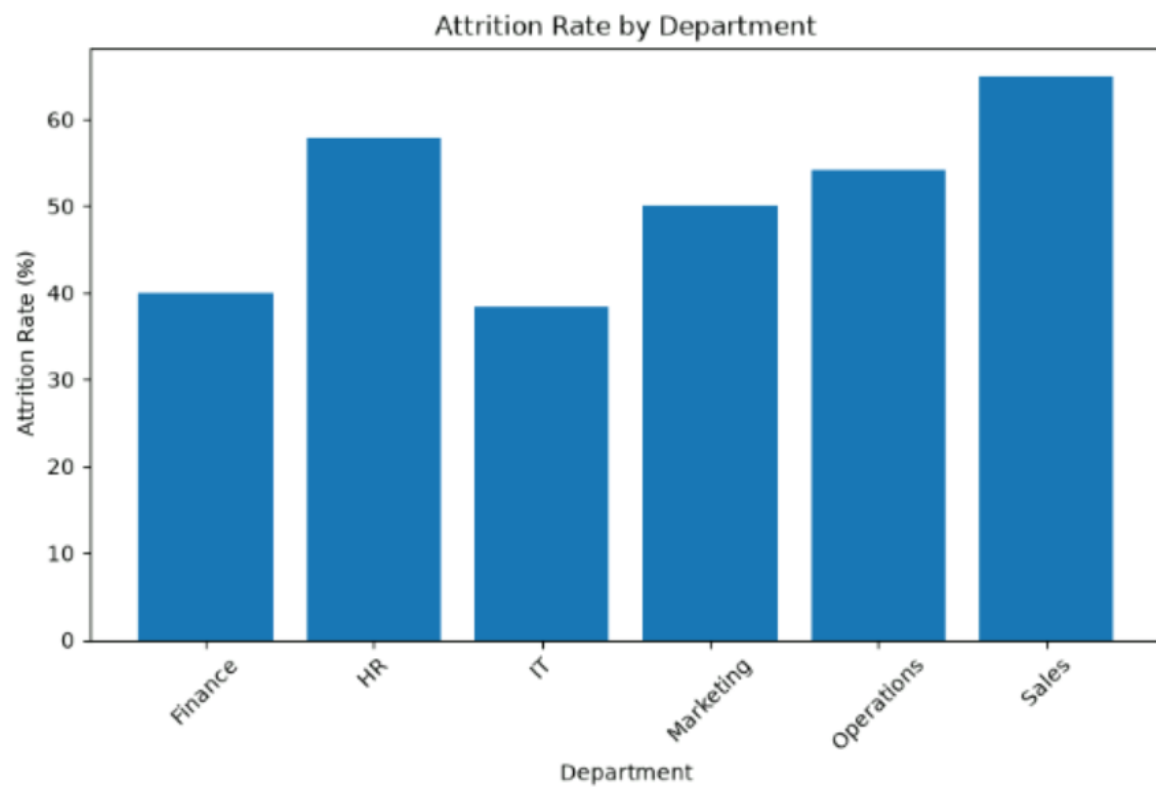
```
plt.show()
```



Work Life Distribution

For rest of Analysis refer git Python Script (Post Analysis)

```
plt.show()
```



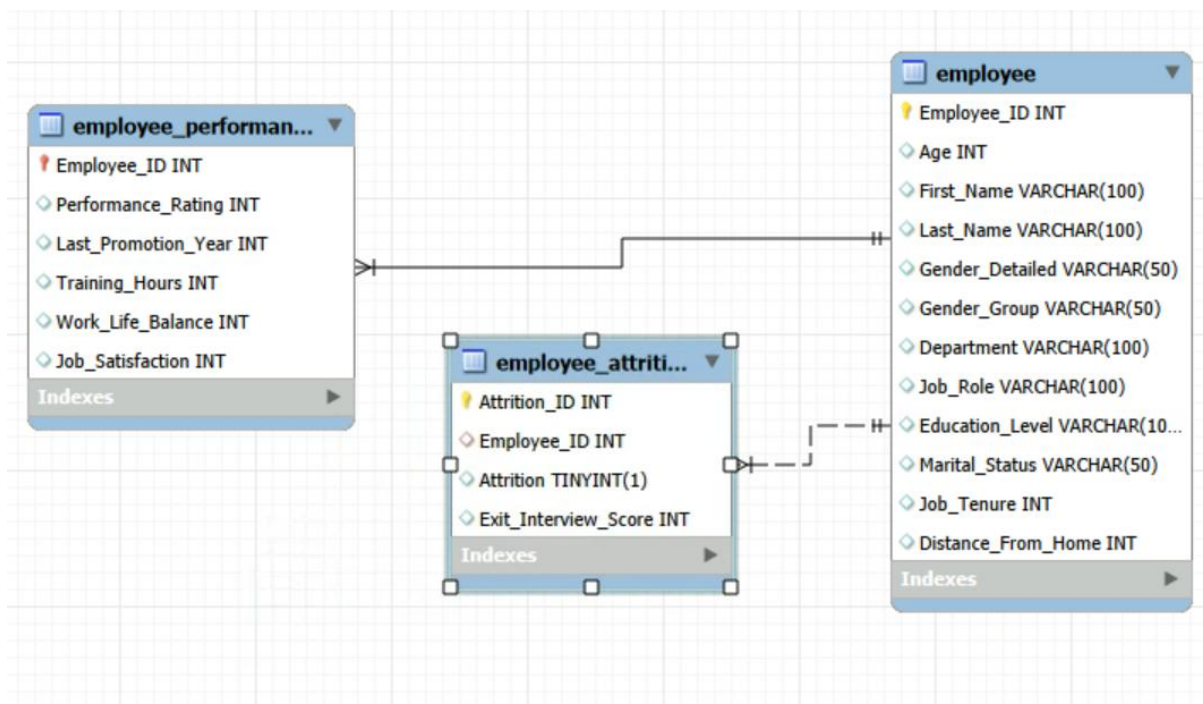
compare employee-level attrition rates across departments with available attrition records.

CHAPTER 4: MYSQL DATA MODELING & IMPLEMENTATION

4.1: Chapter objective

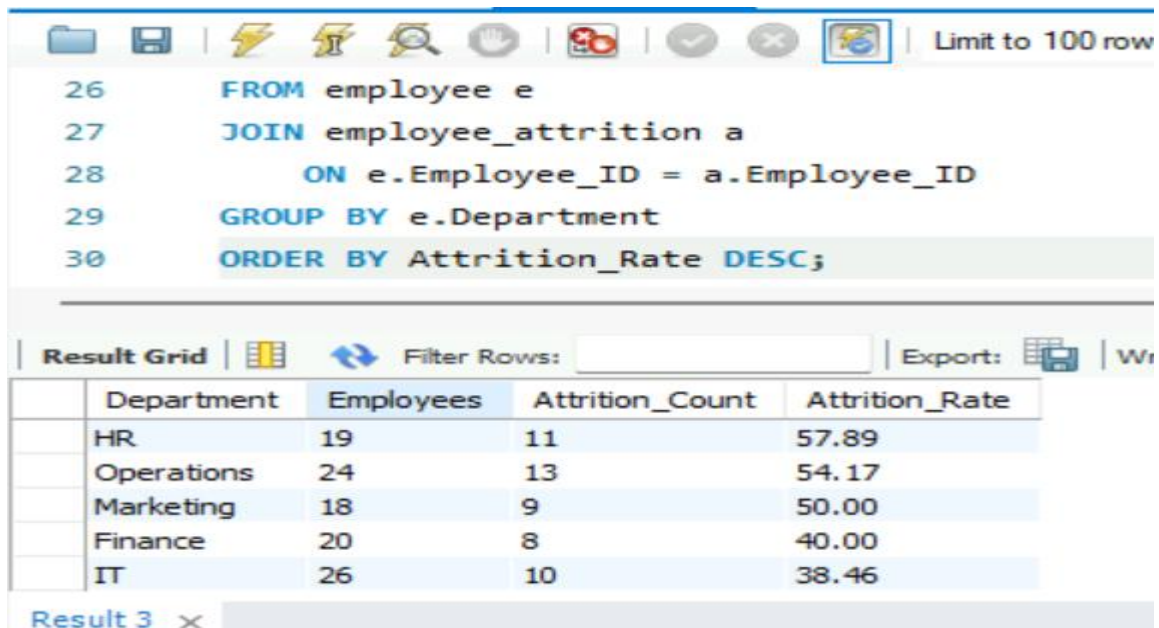
To create a relational database model, import the cleaned datasets into MySQL, establish table relationships and perform selected SQL-based analysis.

4.2: Table and Model Creation



MySQL Relational Data Model

SQL Analysis 1: Attrition by Department



The screenshot shows a SQL query window with the following query:

```
26 FROM employee e
27 JOIN employee_attrition a
28     ON e.Employee_ID = a.Employee_ID
29 GROUP BY e.Department
30 ORDER BY Attrition_Rate DESC;
```

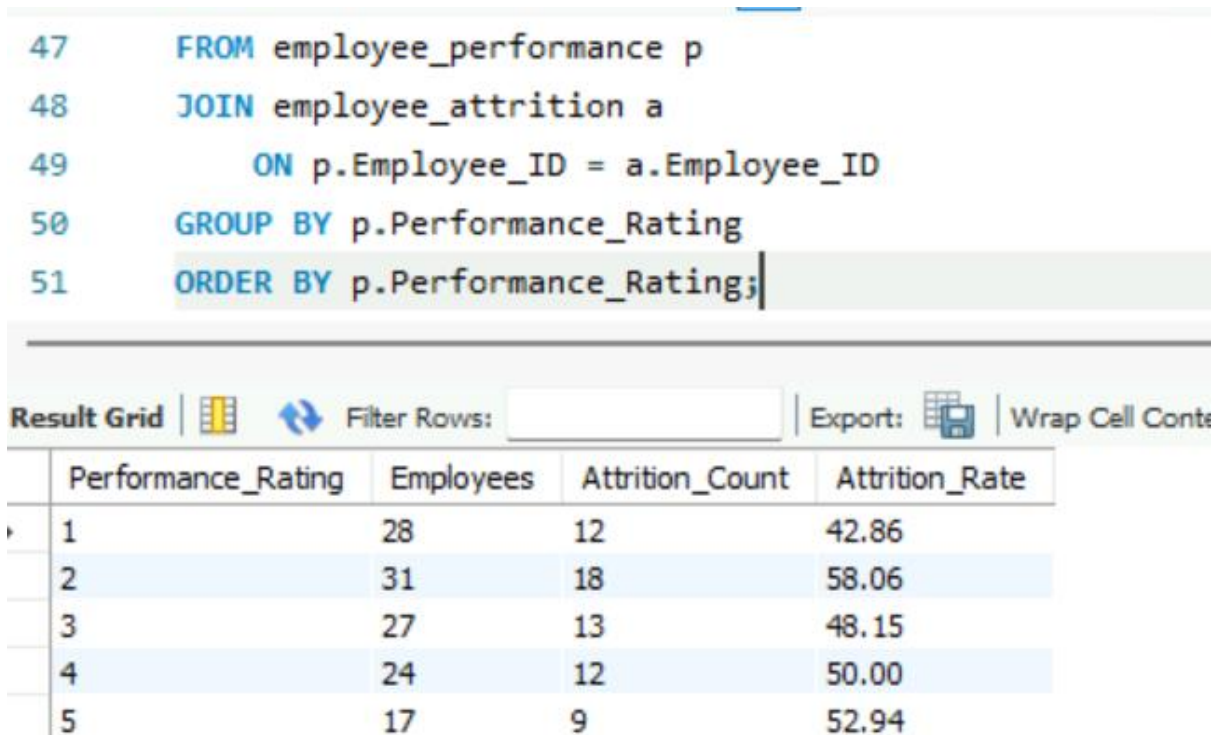
Below the query, the 'Result Grid' tab is active, displaying a table with 5 columns: Department, Employees, Attrition_Count, and Attrition_Rate. The table contains 5 rows of data, sorted by Attrition_Rate in descending order. The interface also includes a 'Filter Rows' field and an 'Export' button.

Department	Employees	Attrition_Count	Attrition_Rate
HR	19	11	57.89
Operations	24	13	54.17
Marketing	18	9	50.00
Finance	20	8	40.00
IT	26	10	38.46

Result 3 x

Department Attrition Analysis: SQL was used to calculate department-wise employee attrition rates from the relational database.

SQL Analysis 2: Attrition by Performance Rating



The screenshot shows a SQL query window with the following query:

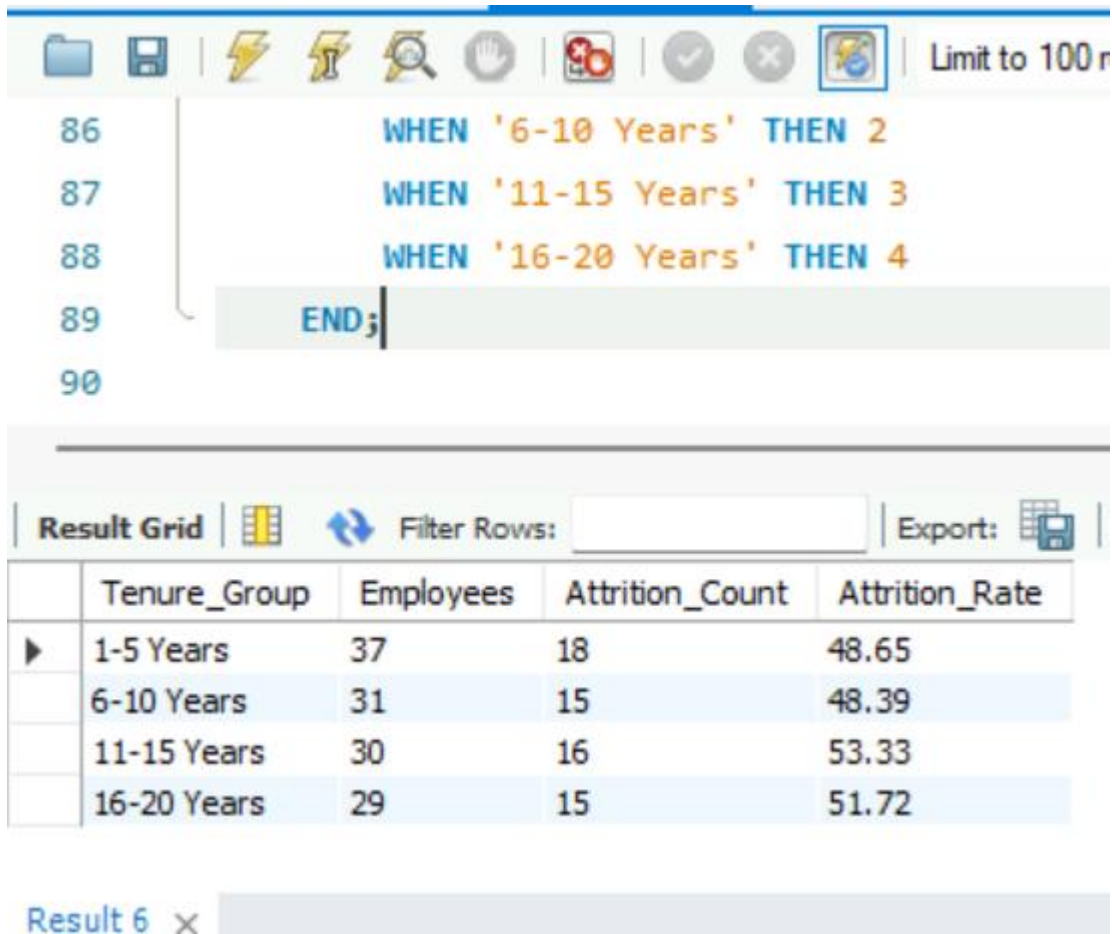
```
47 FROM employee_performance p
48 JOIN employee_attrition a
49     ON p.Employee_ID = a.Employee_ID
50 GROUP BY p.Performance_Rating
51 ORDER BY p.Performance_Rating;
```

Below the query, the 'Result Grid' tab is active, displaying a table with 5 columns: Performance_Rating, Employees, Attrition_Count, and Attrition_Rate. The table contains 5 rows of data, sorted by Performance_Rating in ascending order. The interface also includes a 'Filter Rows' field and an 'Export' button.

Performance_Rating	Employees	Attrition_Count	Attrition_Rate
1	28	12	42.86
2	31	18	58.06
3	27	13	48.15
4	24	12	50.00
5	17	9	52.94

Performance Attrition Analysis: SQL was used to compare employee-level attrition rates across performance ratings.

SQL Analysis 3: Attrition by Job Tenure



The screenshot displays a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search. The SQL editor shows a CASE WHEN statement with four conditions for job tenure groups. The results are displayed in a 'Result Grid' below the editor. The grid has five columns: 'Tenure_Group', 'Employees', 'Attrition_Count', and 'Attrition_Rate'. The data is as follows:

	Tenure_Group	Employees	Attrition_Count	Attrition_Rate
▶	1-5 Years	37	18	48.65
	6-10 Years	31	15	48.39
	11-15 Years	30	16	53.33
	16-20 Years	29	15	51.72

Below the grid, there is a tab labeled 'Result 6' with a close button (x).

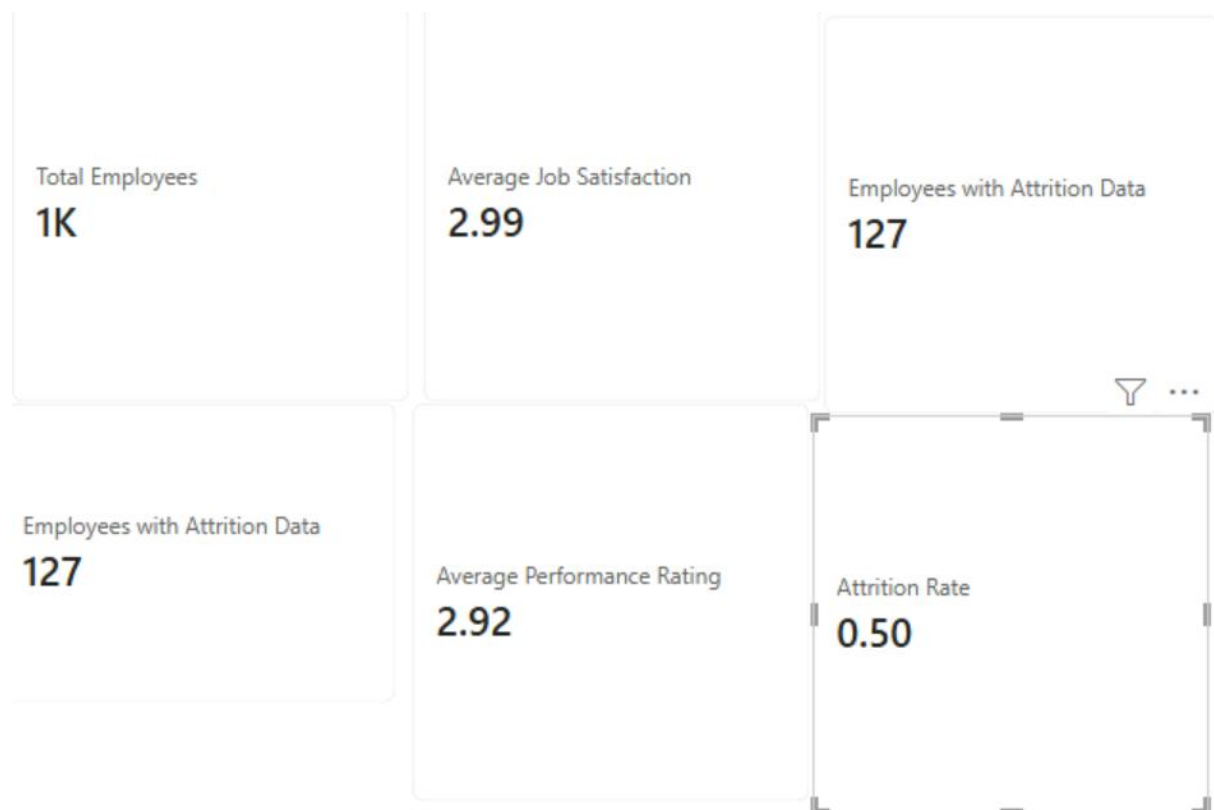
Tenure Attrition Analysis: SQL was used to calculate employee-level attrition rates across defined job-tenure groups.

Chapter 5: Power BI Dashboard Implementation

5.1: Connect PowerBI to DB

Connected

5.2: Create Some Important KPI measures



5.3: Final PowerBI Dashboard.



5.4: Dashboard Output and Key Findings

KPI	Output
Total Employees	1,000
Employees with Attrition Data	127
Attrition Count	64
Attrition Rate	50.39%
Average Job Satisfaction	2.99 / 5
Average Performance Rating	2.92 / 5

5.5: Key Dashboard Findings

- The dataset contains 1,000 employees, while attrition information is available for 127 employees.
 - 64 employees are identified with attrition, resulting in an employee-level attrition rate of 50.39% among employees having attrition records.
 - Sales has the highest observed department-level attrition rate at 65.00%, followed by HR at 57.89%.
 - The observed attrition rate varies across performance ratings, with Rating 2 showing 58.06%.
 - Employees in the 11–15 year tenure group show an attrition rate of 53.33%.
 - Attrition rates vary across job satisfaction levels, with the highest observed rate at 54.55% for Satisfaction Level 3.
 - Attrition rates also vary across work-life balance ratings, with the highest observed rate of 57.14% at ratings 1 and 3.
 - The average Job Satisfaction is 2.99/5, while the average Performance Rating is 2.92/5.
-

https://github.com/amankumar1ext-web/Case_Study/tree/main/W2_Case_Study_1